

Tutorial 11

CS105.3 Database Management Systems

Question 01

Normalization

General introduction

Imagine we're building a restaurant management application. That application needs to store data about the company's employees, and it starts out by creating the following table of employees:

Employee Unnormalized Table

EMPLOYEE_ID	NAME	CONTACT NO.	JOB_CODE	JOB	STATE_CODE	HOME_STATE
E001	Alice	0773453453/ 0772342342	J01	Chef	36	Galle
E002	Catherin	0773453453/ 0772349342	J02 J04	Waiter /Helper	26	Matara
E003	Bob	0776663457	J02	Waiter	46	Kegalle
E004	Nike	0773553453	J03 J04	Cleaner / Helper	56	Colombo
E005	Shevin	077233453	J04	Helper	66	Negombo
E006	Michel	0773553453	J01	Chef	21	Nugegoda

Illustrate the process of normalizing the data shown in this table to First normal form (1NF), Second Normal Form (2NF) and Third Normal Form (3NF).

What is normalization?

- Normalization is the process for assigning attributes to entities. Properly executed, the normalization process eliminates uncontrolled data redundancies, thus eliminating the data anomalies and the data integrity problems that are produced by such redundancies. Normalization does not eliminate data redundancy; instead, it produces carefully controlled redundancy that lets us properly link database tables.

When is the table in 1NF?

- A table is in 1NF when all the key attributes are defined (no repeating groups in the table) and when all remaining attributes are dependent on the primary key. However, a table in 1NF still may contain partial dependencies, i.e., dependencies based on only part of the primary key.

When is a table in 2NF?

- A table is in 2NF when it is in 1NF and it includes no partial dependencies. However, a table in 2NF may still have transitive dependencies, i.e., dependencies based on attributes that are not part of the primary key.

When is a table in 3NF?

A table is in 3NF when it is in 2NF and it contains no transitive dependencies.

When is the table in BCNF?

A table is in Boyce-Codd Normal Form (BCNF) when it is in 3NF and every determinant in the table is a candidate key. For example, if the table is in 3NF and it contains a nonprime attribute that determines a prime attribute, the BCNF requirements are not met.

This description clearly yields the following conclusions:

- If a table is in 3NF and it contains only one candidate key, 3NF and BCNF are equivalent.
- BCNF can be violated only if the table contains more than one candidate key. Putting it another way, there is no way that the BCNF requirement can be violated if there is only one candidate key.